



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-09029

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Wed, May 1, 2024 at 10:25 PM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

02-May-2024

Dear Dr. Khare:

The manuscript entitled "Identification of Non-Speaking and Minimal-Speaking Individuals Using Nonverbal Vocalizations" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott

Editor-in-Chief, IEEE Access
ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Identification of Non-Speaking and Minimal-Speaking Individuals Using Nonverbal Vocalizations

AUTHORS: Tran, Van-Thuan; Tsai, Wei-Ho

ABSTRACT: Speech remains a prevalent mode of communication powering various intelligent functions in human-computer interaction applications, notably in Speaker/Person Identification (PID) systems. However, there is a considerable population of Non-speaking and Minimal-speaking (NMS) individuals, who heavily rely on nonverbal vocalizations for communication, and the existing speech-based PID systems may not be suitable for users from this community. This study delves into the use of nonverbal vocalizations to identify NMS subjects, termed as NMS-PID, and explores the feasibility of developing an identification system, namely S-NMS-PID, that accommodates both speaking users (with speech input) and NMS users (with nonverbal-vocalization input). Leveraging the recently published ReCANVo dataset of NMS nonverbal vocalizations and our speech dataset, our experiments with multiple networks and acoustic features demonstrate promising results for NMS-PID and S-NMS-PID, evident in average accuracies ranging from 70% to 92%. The proposed convolutional recurrent neural network-based model, despite its smaller size, achieves results nearly on par with much deeper models such as VGG16 and ResNet50. Our findings also highlight the efficacy of Mel-frequency cepstral coefficients features compared to the spectrogram features. Furthermore, a two-step training strategy involving supervised contrastive learning for representation learning followed by fine-tuning with cross-entropy loss significantly enhances robustness and accuracy, particularly in classifying data from minority classes, enhancing overall performance. This study's outcomes hold potential for tailoring human-computer interaction applications specifically for NMS users. Implementing NMS-PID and S-NMS-PID in security and authentication processes ensures secure and reliable user identification across diverse platforms, transcending sole reliance on speech-based methods.

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If you have any questions, please contact the article administrator: Dr. Manpreet Kaur m.kaur@ieee.org. In addition, if you would like to get a certificate after completing this review, please send an email to ieeeaccess@ieee.org.

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